10.2 Annex 2: Request-based Pricing Formula

The formula assumes there is a minimal unit of work to measure network cost, called a request (comparable to opcode). Taking the minimum time-unit of service s (seconds), the capacity of the network C_{net}, and the cost of the network per such time-unit, $Cost_{net}$ the price of 1 request per second is:

$$P_{base} = Cost_{net}/C_{net}$$

To maintain a buffer, define $C_{max} < C_{net}$ as the maximum capacity for sale. The adjusted base price for break-even is:

$$P_{base} = Cost_{net}/C_{max}$$

For a given amount $r < C_{max}$ of requests per second, using a subsidy-factor $s \leq 1$:

$$P(r) = P_{base} \cdot (1 - s) \cdot r$$

This allows for fixed pricing (e.g., 0.0001$ per request) while knowing the subsidy amount.

To make pricing more dynamic, four factors are considered:

1. Available capacity (f_a): Increases price as network utilization increases.
2. Timeslot (f_b): For when the rate-limit is supposed to start.
3. Timerange (f_c): Time until expiry.
4. Requested capacity (f_d): Decreases price per request for larger capacity requests (bulk discount).

Each factor is modeled as a value between 1 and $f_{a_{max}}$ (which can differ for each factor).

The aggregated pricing formula using these factors is:

$$P(f_a, f_b, f_c, f_d)(r) = P_{base} \cdot (1 - s) \cdot (f_a \cdot f_b \cdot f_c \cdot f_d)^{1/4} \cdot r$$